Graph Attention Networks for Drug-Target Interaction Prediction

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Preprint, 2023
Synthetic demo paper generated for ResearchMind AI testing. Not a real publication.

Abstract

Predicting drug-target interactions is a central step in computational drug discovery. We propose DTI-GAT, a model that applies a Graph Attention Network to molecular graphs and combines it with a convolutional neural network over protein sequences. We show that attention over molecular substructures improves interaction prediction. On the DAVIS dataset, DTI-GAT achieves an AUC of 0.91, outperforming the DeepDTA baseline by 4%.

1 Introduction

Drug discovery pipelines rely on screening large compound libraries against protein targets. Existing deep learning models such as DeepDTA encode drugs as strings and ignore molecular topology. We argue that graph-structured representations of molecules capture binding-relevant chemistry better than string encodings.

2 Method

DTI-GAT represents each drug as a molecular graph where atoms are nodes and bonds are edges. A Graph Attention Network with four attention heads produces atom embeddings, which are pooled into a drug vector. Protein sequences are encoded with a convolutional neural network. The two vectors are concatenated and passed to a multilayer perceptron.

3 Experiments

We evaluate on the DAVIS dataset, which contains kinase inhibitor binding measurements, and on BindingDB. We use a random split with 80% training, 10% validation and 10% test data.

Results show that DTI-GAT achieves an AUC of 0.91 on DAVIS and 0.88 on BindingDB. Attention-based graph models generalize well to unseen proteins in our experiments, improving accuracy by 3% on held-out targets.

4 Limitations

Our evaluation is limited to kinase targets from DAVIS and BindingDB. We did not evaluate a cold-start setting in which test proteins are completely unseen during training. The model ignores three-dimensional protein structure.

Future work should evaluate DTI-GAT on the KIBA dataset and on cold-start protein splits.

5 Conclusion

We presented DTI-GAT, demonstrating that graph attention over molecular structure improves drug-target interaction prediction.

References

[1] Placeholder reference for the synthetic paper.